

Daniel Dong

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WORK EXPERIENCE

- Software Developer** | *Samaritan Ministries International* | Peoria, Illinois July 2024 - Present
- Engineered a **recursive C# algorithm** distributing **\$2M/month** across a \$20M+/year bill-sharing program, preventing billing spikes and reducing 300+ support calls monthly
 - Built an **automated Bill-Sharing service** processing **\$200K/month** and 1,000+ bills per run, eliminating 50+ hours of monthly work for Processors
 - Delivered production updates and features to a large-scale **React** application serving 250K+ members for bill management, share tracking, and real-time in-app messaging
- Software Engineer Intern** | *Suitcase* | Ann Arbor, MI May 2023 - August 2023
- Architected the full-stack solution for Suitcase, a banquet room visualization tool, enabling dynamic 3D asset management and real-time user interactions
 - Developed **Django backend APIs** to serve the Unity frontend, managing user, vendor, and product data efficiently
 - Optimized a **PostgreSQL** database and asset delivery via **AWS S3/CloudFront**, achieving 60% faster room loading times for 5000+ 3D models
- Software Engineer Intern** | *Altair Engineering* | Irvine, California May 2022 - August 2022
- Cleaned up thousands of Fortran77 and Fortran90 files by developing automated Python scripts to reduce errors, enhance codebase cleanliness, and increase development efficiency
 - Streamlined complex Fortran declarations/subroutines, replaced deprecated functions, and removed unused variables with scripts, earning recognition from the HyperWorks Core Development team

TECHNICAL SKILLS

Languages: Python, C#, JavaScript, SQL, Kotlin, C++, TypeScript, HTML/CSS, Java
Frameworks: .NET Core, Entity Framework, Django, React, Vue.js, Node.js
Tools/Platforms: Git, AWS (EC2, S3, RDS, CloudFront), PostgreSQL, Supabase, SQLite, Linux, Visual Studio

RELEVANT PROJECTS

- Reminisce - Travel Scrapbook App** | *Kotlin & Python*
- Developed a **Django**-based backend with **RESTful APIs**, integrating Google Maps, Spotify, and Weather APIs for improved user engagement with real-time route and weather trackers and music preferences
 - Optimized data exchange between a **Kotlin-based mobile application** and its backend by refining JSON structures and API endpoints, enhancing data retrieval speeds and enabling real-time updates for 100+ data points per user
- RANKED - Offline-First Ranking App** | *React Native, TypeScript*
- Built a cross-platform mobile app for creating and managing custom rankings using **React Native**, with **Supabase** for authentication and cloud synchronization
 - Implemented an **offline-first architecture** with local **SQLite** storage as the primary data source and background sync for remote updates and conflict resolution
 - Ensured data consistency through **transaction-safe operations**, rank offset handling, and optimistic updates with automatic rollback on failure
- Multi-Threaded Network Fileserver** | *C++*
- Engineered a network file server with socket programming for efficient client-server communication with multiple clients
 - Implemented multi-threading to manage concurrent client requests with thread synchronization for high server responsiveness and effective resource allocation
 - Developed read, write, create, & delete file system functionalities with crash consistent protocols to ensure data integrity

EDUCATION

University of Michigan - Ann Arbor April 2024
B.S. in Computer Science and Cognitive Science

Relevant Coursework: Data Structures and Algorithms, Web Systems, Operating Systems, Database Management Systems, Machine Learning, Mobile App Development, Computer Vision

Interests: Sports (Basketball, Tennis, Volleyball, NBA), Music (Vocals, Guitar, Drums, Songwriting), Cooking, Baking
Fluent Languages: English, Mandarin, French